RAG Learning

Module B2 — Fundamental Data Analysis

2.1 Physical Quantities and SI Units

Physical quantity: A property of an object or phenomenon that can be measured and expressed as a number multiplied by a unit. Every physical quantity has both a magnitude and a unit.

Physics is built on a system of well-defined quantities and units. The International System of Units (SI) provides a coherent framework for all measurements in science. There are seven base quantities, each with a corresponding base unit.

The Seven SI Base Units

Base Quantity	Symbol	SI Base Unit	Unit Symbol
Mass	m	kilogram	kg
Length	l	metre	m
Time	t	second	s
Electric current	I	ampere	A
Temperature	T	kelvin	K
Amount of substance	n	mole	mol
Luminous intensity	I_v	candela	cd

Derived Units

All other physical quantities are derived from combinations of the base units. Some derived units are given special names for convenience.

Quantity	Derived Unit	In Base Units
Force	newton (N)	$kg\ m\ s^{-2}$
Energy / Work	joule (J)	$kg\ m^2\ s^{-2}$
Power	watt (W)	$kg\ m^2\ s^{-3}$
Pressure	pascal (Pa)	$kg\ m^{-1}\ s^{-2}$
Charge	coulomb (C)	$A\ s$
Potential difference	volt (V)	$kg\ m^2\ s^{-3}\ A^{-1}$
Resistance	ohm (Ω)	$kg\ m^2\ s^{-3}\ A^{-2}$
Frequency	hertz (Hz)	s^{-1}

Exam Tip: Being able to derive units from equations is a commonly tested skill. For example, from F = ma, the unit of force is $kg \times m\ s^{-2} = kg\ m\ s^{-2} = N$. You should be able to work backwards too: given units, deduce the equation.

SI Prefixes

SI prefixes are used to express very large or very small quantities conveniently.

Prefix	Symbol	Factor
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}

kilo	k	10^3
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

2.2 Scalars and Vectors

Scalar: A quantity that has magnitude only. Examples: mass, speed, energy, temperature, distance, time.

Vector: A quantity that has both magnitude and direction. Examples: force, velocity, acceleration, displacement, momentum.

Vectors can be represented by arrows, where the length represents the magnitude and the arrow shows the direction. Vectors can be added using the triangle of vectors or parallelogram rule. For perpendicular vectors, use Pythagoras' theorem for the resultant magnitude and trigonometry for the direction.

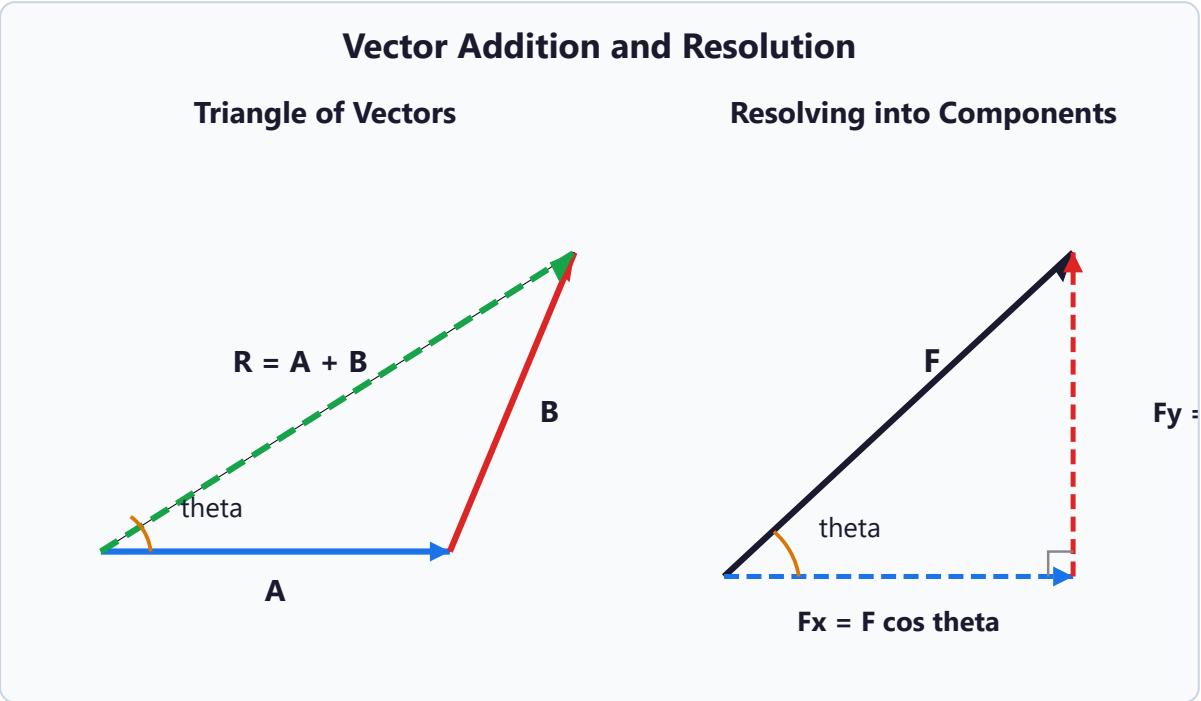
Resolving Vectors into Components

Any vector can be resolved into two perpendicular components. For a vector of magnitude F at angle θ to the horizontal:

$$\text{Horizontal component: } F_x = F \cos \theta$$

$$\text{Vertical component: } F_y = F \sin \theta$$

This technique is essential for analysing forces on inclined planes, projectile motion, and many other situations.



2.3 Estimation and Orders of Magnitude

Order of magnitude: The power of ten closest to the value of a quantity. Two quantities are said to be of the same order of magnitude if they are within a factor of about 10 of each other.

Estimation is a key physics skill. You should be able to make reasonable estimates by combining known quantities. OCR B particularly values this skill and frequently tests it.

Useful Estimates to Know

Quantity	Approximate Value
Mass of an adult human	70 kg
Height of a person	1.7 m

Walking speed	$1.5\ m\ s^{-1}$
Speed of sound in air	$340\ m\ s^{-1}$
Speed of light	$3 \times 10^{8}\ m\ s^{-1}$
Atmospheric pressure	1×10^{5} Pa
Density of water	$1000\ kg\ m^{-3}$
Acceleration due to gravity (g)	$9.81\ m\ s^{-2}$
Diameter of an atom	$\sim 10^{-10}$ m
Diameter of a nucleus	$\sim 10^{-15}$ m
Mass of an electron	9.11×10^{-31} kg
Mass of a proton/neutron	1.67×10^{-27} kg
Charge of an electron	1.60×10^{-19} C

Exam Tip: When making estimates, show your working clearly and state any assumptions. Examiners are looking for a sensible approach and a reasonable order of magnitude, not an exact answer. Break the problem into steps using quantities you know.

2.4 Homogeneity of Equations

Homogeneous equation: An equation in which every term has the same SI base units. A physically valid equation must be homogeneous, though a homogeneous equation is not necessarily physically correct (numerical constants and dimensionless factors cannot be checked).

Checking homogeneity (dimensional analysis) is a powerful technique for verifying equations, deriving possible relationships, and spotting errors. To check homogeneity, substitute the base units for each

quantity and verify that both sides of the equation reduce to the same combination of base units.

Worked Example

Check the homogeneity of the kinetic energy equation: $E_k = ½mv^2$

- Left side: E_k has units of J = kg m^2 s^{-2}
- Right side: m has units kg, v^2 has units $(m\ s^{-1})^2 = m^2\ s^{-2}$
- So the right side gives kg × m^2 s^{-2} = kg m^2 s^{-2}
- Both sides match, so the equation is homogeneous.

2.5 Graphical Methods

Graphs are central to data analysis in OCR B Physics. Understanding how to use graphs to determine relationships and extract physical quantities is essential.

Straight-Line Graphs

The equation of a straight line is y = mx + c, where m is the gradient and c is the y-intercept. If experimental data produces a straight line when plotted, the relationship between the variables is linear.

Linearisation of Non-Linear Relationships

Many physical relationships are not linear, but they can be transformed into a straight-line form by choosing appropriate variables to plot.

Relationship	Plot	Gradient	Y-intercept
$y = kx^2$	y vs x^2	k	0
y = k/x	y vs 1/x	k	0
$y = kx^n$	log y vs log x	n	log k
$y = Ae^{kx}$	ln y vs x	k	ln A

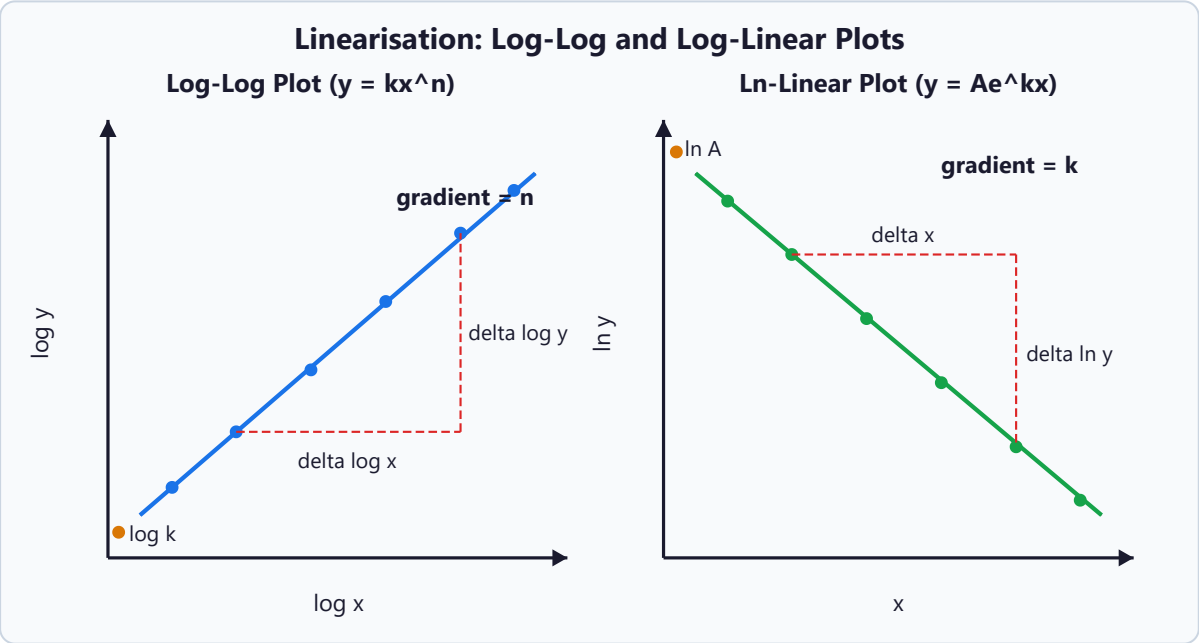
$y = Ae^{-x/\tau}$	ln y vs x	$-1/\tau$	ln A
$T^2 = (4\pi^2/g)l$	T^2 vs l	$4\pi^2/g$	0

Exam Tip: OCR B frequently asks you to suggest what to plot to obtain a straight line. Always rearrange the equation into y = mx + c form first. Whatever is multiplied by the independent variable becomes the gradient. Whatever is left over becomes the intercept.

Using Logarithmic Plots

If you suspect a power law ($y = kx^n$) but do not know the power n, plot log y against log x. The gradient gives n and the y-intercept gives log k. This is because:

$$\log y = n \log x + \log k$$



For exponential relationships ($y = Ae^{kx}$), plot ln y against x. The gradient gives k and the y-intercept gives ln A:

$$\ln y = kx + \ln A$$

Area Under Graphs

The area under a graph often has physical significance. Common examples include:

- Area under a velocity-time graph = displacement
- Area under a force-displacement graph = work done (energy transferred)
- Area under a force-time graph = impulse (change in momentum)
- Area under a power-time graph = energy transferred
- Area under a current-time graph = charge

2.6 Significant Figures and Standard Form

Rules for Significant Figures

- All non-zero digits are significant
- Zeros between non-zero digits are significant (e.g., 305 has 3 s.f.)
- Leading zeros are not significant (e.g., 0.0042 has 2 s.f.)
- Trailing zeros after a decimal point are significant (e.g., 2.50 has 3 s.f.)
- Trailing zeros before a decimal point may or may not be significant (use standard form to clarify)

Standard Form

Standard form (scientific notation) expresses numbers as $A \times 10^n$ where $1 \leq A < 10$. This removes ambiguity about significant figures and makes it easier to work with very large or very small numbers. For example, $0.00340 = 3.40 \times 10^{-3}$ (3 s.f.).

Exam Tip: When performing calculations, carry extra significant figures through intermediate steps and only round the final answer. State your final answer to an appropriate number of significant figures, typically matching the least precise value used in the calculation.

2.7 Mathematical Skills for Physics

OCR B expects competence in a range of mathematical techniques applied to physics problems:

- **Rearranging equations:** Isolate the desired variable by performing inverse operations on both sides

- **Trigonometry:** sin, cos, tan and their inverses for resolving vectors and analysing wave behaviour
- **Pythagoras' theorem:** For finding resultant vectors and distances
- **Logarithms and exponentials:** Particularly for radioactive decay, capacitor discharge, and data analysis
- **Differentiation (conceptual):** The gradient of a displacement-time graph gives velocity; the gradient of a velocity-time graph gives acceleration
- **Integration (conceptual):** The area under a graph represents the integral of the y-quantity with respect to the x-quantity

Natural logarithm (ln): The logarithm to base e (where e ≈ 2.718). If $y = e^x$, then $x = \ln y$. Used frequently in exponential decay and growth problems in physics.

Worked Example: An experiment gives the relationship $T = kL^n$. Data: when L = 0.50 m, T = 1.42 s; when L = 2.00 m, T = 2.84 s. Find n and k.

Taking logs: $\log T = n \log L + \log k$

Point 1: $\log(1.42) = n \log(0.50) + \log k \Rightarrow 0.152 = -0.301n + \log k$
Point 2: $\log(2.84) = n \log(2.00) + \log k \Rightarrow 0.453 = 0.301n + \log k$

Subtracting: $0.453 - 0.152 = 0.301n - (-0.301n) = 0.602n$
$0.301 = 0.602n$, so n = **0.50** (i.e., $T \propto \sqrt{L}$, consistent with a pendulum: $T = 2\pi\sqrt{(L/g)}$)

$\log k = 0.152 + 0.301(0.50) = 0.303$, so $k = 10^{0.303}$ = **2.01**
This gives $k = 2\pi/\sqrt{g} = 2\pi/\sqrt{9.81} = 2.01$ ✓

Worked Example: Estimate the number of tennis balls that would fill a classroom.

Classroom dimensions: approximately 10 m × 8 m × 3 m = 240 m^3
Tennis ball diameter: approximately 6.5 cm = 0.065 m
Volume of one ball: $(4/3)\pi r^3 = (4/3)\pi(0.0325)^3 = 1.44 \times 10^{-4}$ m^3
Packing efficiency (random packing): approximately 64%
Number of balls = $0.64 \times 240 / 1.44 \times 10^{-4} \approx$ **1.1×10^6** ≈ 1 million balls

Order of magnitude: **10^6**. The key is showing clear working and stating assumptions.